

Lab 12

Introduction to MS Access and Creating a Simple Database

Objective:

1. To understand the basic concepts of database management systems.
2. To learn the components of MS Access such as tables, fields, records, queries, forms, and reports.
3. To create a simple database using MS Access.
4. To enter, edit, and manage data in tables.
5. To generate queries, forms, and reports for better data organization and presentation.

Theory

What is MS Access?

Microsoft Access is a **Relational Database Management System (RDBMS)** developed by Microsoft that combines the power of a database with a user-friendly interface. It allows users to store large amounts of information, organize it systematically, and retrieve it efficiently.

Key Components of MS Access

- **Database:** A collection of related data stored in an organized manner.
- **Table:** The basic unit of storage that consists of rows (records) and columns (fields).
- **Field:** A single piece of data or attribute of a record (e.g., Name, Roll No).
- **Record:** A complete set of fields representing one entry.
- **Primary Key:** A unique identifier for each record in a table.
- **Query:** A way to search or filter data based on conditions.
- **Form:** A user-friendly interface to enter or view data.
- **Report:** A formatted output of data for printing or presentation.

Advantages of MS Access

- Easy to create and manage databases
- Supports relational data
- Provides built-in templates
- Useful for small to medium-sized applications
- Integrates with other Microsoft Office tools

Requirements

- Computer System
- Microsoft Access Software
- Basic Computer Knowledge

Database File:

It is a file which stores the entire database. The database file is saved to your hard drive or other storage devices.

Datatypes:

Datatypes are the properties of each field. Every field has one datatype like text, number, date, etc.

Table

- A Table is an object which stores data in Row & Column format to store data.
- A Table is usually related to other tables in the database file.
- Each column must have Unique name
- We can also define Primary Key in a table.

Query

- Queries answer a question by selecting and sorting and filtering data based on search criteria.
- Queries show a selection of data based on criteria (limitations) you provide.
- Queries can pull from one or more related Tables and other Queries.
- Types of Query can be SELECT, INSERT, UPDATE, DELETE.

Form

- A form is a database object that you can use to create a user interface for a database application.
- Forms help you to display live data from the table. It mainly used to ease the process of data entry or editing.

Report

- A report is an object in desktop databases primarily used for formatting, calculating, printing, and summarizing selected data.
- You can even customize the report's look and feel.

MS Access Datatypes

MS Access common data types are listed below:

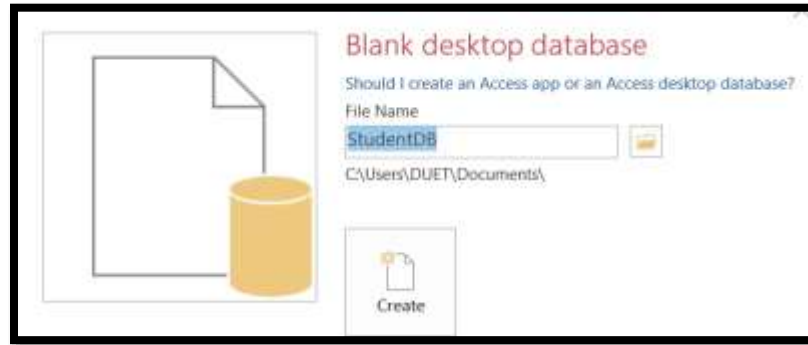
Type of Data	Description
Short Text	Text, including numbers which does not need calculation. (e.g., Mobile numbers).
Long Text	This data type is used for lengthy text or alphanumeric data.
Number	Numeric data type used for storing mathematical calculations.
Date/Time	Store Date/time for the years 100 through 9999.
Currency	It allows you to store currency values and numeric data with one to four decimal places.
Auto Number	Assign a unique number or assigned by Microsoft Access when any new record is created. Usually used as the primary key
Yes/No	It only stores logical values Yes and No.
Attachment	It stores files, such as digital photos. Multiple files can be attached per record.
OLE objects	OLE objects can store audio, video, other Binary Large Objects.
Hyperlink	Text or combinations of text and numbers stored. That text is used as hyperlink address.
Calculated	Helps you to create an expression that uses data from one or more fields.

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Procedure

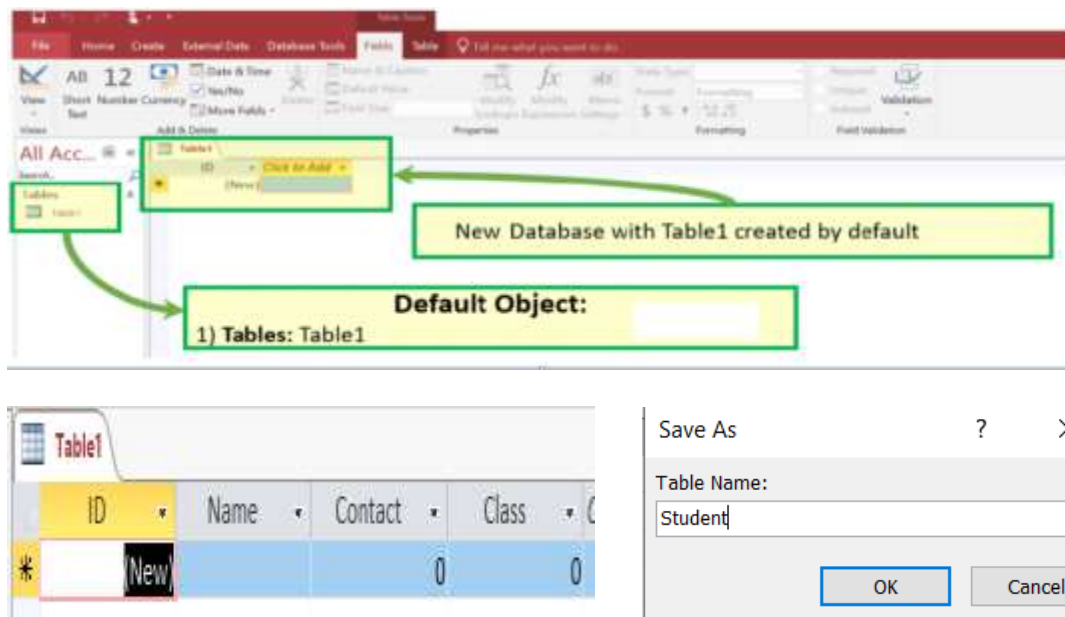
Creating a New Database

1. Open **Microsoft Access**.
2. Choose **Blank Database**.
3. Enter a name for your database (e.g., *StudentDB*).
4. Click **Create**.



Creating a Table

1. Go to **Create** → **Table Design**.
2. Enter field names (e.g., RollNo, Name, Class, ContactNo).
3. Select appropriate data types (Number, Text, Date, etc.).
4. Set **RollNo** as the **Primary Key**.
5. Save the table as **Student**.



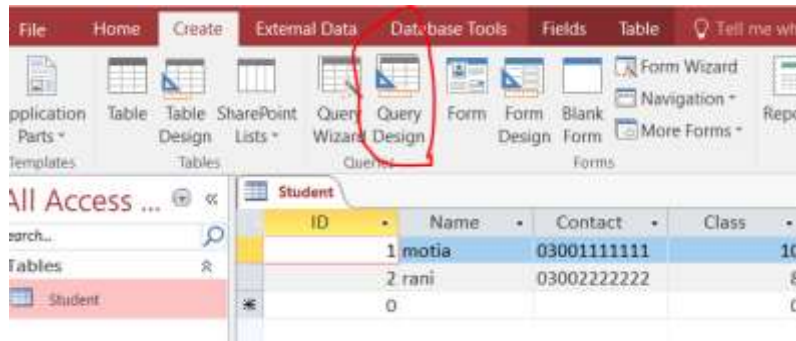
Entering Data

1. Open the table in **Datasheet View**.
2. Type the records manually.
3. Save the table.

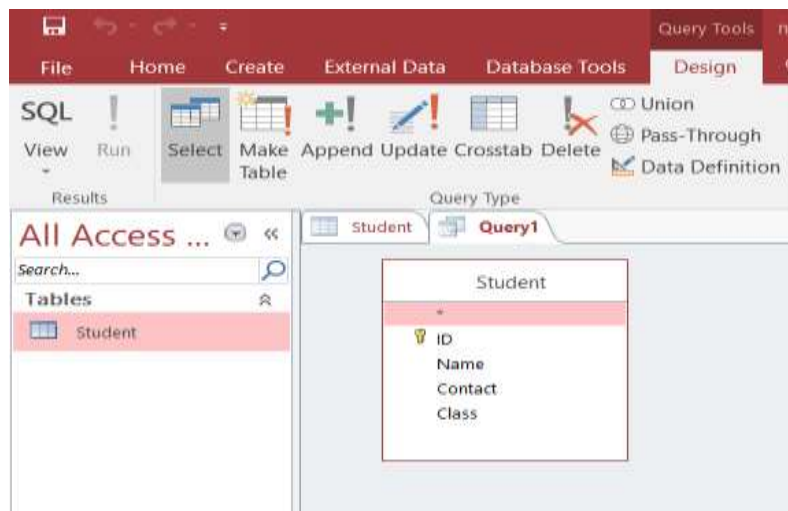
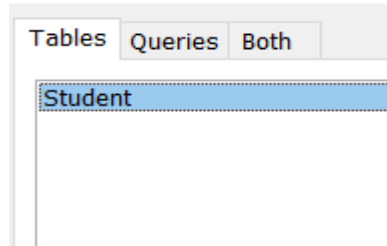
Student				
ID	Name	Contact	Class	Cl
1	motia	03001111111	10	
2	rani	03002222222	8	
*	0		0	

Creating a Query

1. Go to **Create** → **Query Design**.
2. Add the **Student** table.
3. Choose fields you want to include.
4. Specify conditions (e.g., Class = "10").
5. Run the query to view filtered results.
6. Save the query.



Show Table



Creating a Form

1. Select the **Student** table.
2. Click **Create** → **Form**.
3. MS Access automatically generates a form.
4. Save the form.

Creating a Report

1. Select the **Student** table.
2. Click **Create** → **Report**.
3. Modify the layout if needed.
4. Save the report.

Assignment:

1. Create a New Database

- Open **Microsoft Access**.
- Create a **Blank Database** named **StudentDB.accdb**.

2. Create a Table

Create a table named **Student** with the following fields:

Field Name	Data Type	Description
RollNo	Number	Primary Key
Name	Short Text	Student full name
Class	Short Text	Class or Grade
Section	Short Text	Section (A/B/C)
Gender	Short Text	Male/Female
ContactNo	Short Text	Phone Number
DOB	Date/Time	Date of Birth

- Set **RollNo** as the **Primary Key**.
- Save the table.

3. Enter at Least 10 Records

Switch to **Datasheet View** and enter any 10 student records of your choice.

4. Create a Query

Create a query named **ClassQuery** to display all students from a specific class.

Condition:

Show all students where **Class = "10"**.

Save and run the query.